

WEIMAR, Germany

WMO: 105550

Lat: 50.983N

Lon: 11.317E

Elev: 268

StdP: 98.15

Time Zone: 1.00 (EUC)

Period: 97-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.4	-13.4	1.2	-9.1	-10.9	1.5	-7.3	14.4	7.7	13.1	7.1	1.6	80	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	29.9	18.7	27.9	17.8	26.0	17.3	19.7	26.8	19.0	25.5	18.1	23.8	2.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.0	21.1	16.8	12.3	20.5	16.0	11.8	20.1	57.5	26.6	54.9	25.4	52.3	23.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.5	8.0	DB	-14.9	33.7	2.8	1.7	-16.9	34.9	-18.5	35.9	-20.0	36.9	-22.0	38.1	(4)
(5)			WB	-14.6	22.2	3.2	1.3	-16.9	23.1	-18.8	23.9	-20.6	24.6	-22.9	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	0.4	2.2	4.9	8.6	13.6	16.1	17.9	18.1	14.3	9.8	4.8	1.5	(6)	
	DBStd	7.39	5.15	4.80	4.08	3.75	3.33	3.50	3.25	3.10	3.33	4.05	3.70	4.25	(7)	
	HDD10.0	1238	298	221	163	69	7	1	0	0	3	53	158	263	(8)	
	HDD18.3	3360	556	452	416	292	151	84	48	43	126	265	405	521	(9)	
	CDD10.0	1016	1	2	5	27	118	184	246	250	132	47	3	1	(10)	
	CDD18.3	96	0	0	0	0	4	17	36	34	5	0	0	0	(11)	
	CDH23.3	1126	0	0	0	3	87	208	368	379	80	2	0	0	(12)	
	CDH26.7	316	0	0	0	0	16	48	121	121	10	0	0	0	(13)	
(14) Wind	WSAvg	2.9	3.4	4.0	3.3	2.8	2.5	2.4	2.5	2.3	2.5	3.1	2.6	3.2	(14)	
(15) Precipitation	PrecAvg	601	31	30	40	38	63	66	85	65	53	41	49	41	(15)	
	PrecMax	794	67	59	101	99	170	106	149	175	122	99	128	91	(16)	
	PrecMin	395	4	3	5	7	11	14	21	20	10	15	0	10	(17)	
	PrecStd	97	15	15	22	20	32	20	34	33	24	20	26	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	14.4	17.8	23.4	28.5	30.6	32.6	33.1	28.1	22.1	15.3	12.7	(19)	
		MCWB	8.8	9.3	10.9	13.9	17.9	19.3	18.6	18.4	16.9	15.7	11.4	9.7	(20)	
	2%	DB	10.2	12.4	15.4	20.4	25.8	28.0	30.0	29.7	25.9	19.1	12.7	10.6	(21)	
		MCWB	7.3	8.5	9.7	12.6	16.5	18.3	18.6	18.4	16.3	14.3	9.9	8.1	(22)	
	5%	DB	8.9	10.9	13.2	18.0	23.2	25.7	27.5	27.4	23.5	17.2	11.3	8.9	(23)	
		MCWB	6.7	7.8	8.8	11.5	15.4	17.9	18.1	17.4	16.0	13.1	8.9	6.7	(24)	
	10%	DB	7.0	9.1	11.2	15.8	20.6	23.6	24.9	25.1	21.0	15.5	10.0	7.4	(25)	
		MCWB	5.1	6.5	8.0	10.4	14.0	16.6	17.2	17.1	15.2	12.4	8.0	5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.7	11.9	15.1	18.8	21.1	20.7	20.5	18.5	16.4	12.0	10.1	(27)	
		MCDB	12.0	12.8	16.5	21.2	26.7	27.9	27.6	29.0	24.6	20.7	14.0	12.4	(28)	
	2%	WB	7.7	9.0	10.3	13.6	17.2	19.6	19.5	19.6	17.3	14.7	10.3	8.2	(29)	
		MCDB	9.8	12.1	14.0	18.9	23.2	25.9	26.4	27.0	22.9	18.4	12.2	10.4	(30)	
	5%	WB	6.6	8.0	9.3	12.2	16.2	18.5	18.8	18.6	16.4	13.6	9.2	6.8	(31)	
		MCDB	8.9	10.6	12.4	16.5	22.0	24.5	25.0	24.6	21.7	16.4	11.0	8.6	(32)	
	10%	WB	5.0	6.6	8.1	10.9	15.0	17.4	18.0	17.7	15.7	12.7	8.1	5.6	(33)	
		MCDB	6.9	8.9	11.0	14.9	19.7	22.3	23.3	23.1	20.2	15.1	9.8	7.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.0	7.5	10.0	10.8	10.9	10.4	10.8	10.0	7.6	5.3	4.7	(35)	
		MCDBR	6.2	8.0	11.3	14.3	15.8	15.0	15.2	15.7	14.7	10.6	7.1	5.7	(36)	
	5% WB	MCWBR	4.6	5.1	6.7	7.4	7.5	6.8	5.4	5.7	6.4	6.1	4.9	4.5	(37)	
		MCWBR	6.1	6.6	9.7	12.4	13.3	13.8	12.5	12.7	11.8	9.3	6.4	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.326	0.365	0.403	0.407	0.404	0.419	0.412	0.377	0.351	0.327	0.299	(40)	
		taud	2.415	2.376	2.296	2.228	2.246	2.276	2.239	2.281	2.363	2.429	2.436	2.408	(41)	
	Ebn at Noon		726	802	828	834	848	853	831	817	808	762	687	669	(42)	
		Edh at Noon	61	81	106	127	130	127	130	119	98	77	60	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.75	1.46	2.54	4.04	4.87	5.29	4.99	4.29	3.09	1.80	0.84	0.57	(44)	
	RadStd		0.12	0.21	0.32	0.51	0.57	0.45	0.63	0.43	0.42	0.21	0.12	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page